



ProsGradNet: An effective and structured CNN approach for prostate cancer grading from histopathology images

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ABSTRACT

Prostate cancer (PCa) is one of the most prevalent and potentially fatal malignancies affecting men globally. The incidence of prostate cancer is expected to double by 2040, posing significant health challenges. This anticipated increase underscores the urgent need for early and precise diagnosis to facilitate effective treatment and management. Histopathological analysis using Gleason grading system plays a pivotal role in clinical decision making by classifying cancer subtypes based on their cellular characteristics. This paper proposes a novel deep CNN model named as Prostate Grading Network (ProsGradNet), for the automatic grading of PCa from histopathological images. Central to the approach is the novel Context Guided Shared Channel Residual (CGSCR) block, that introduces structured methods for channel splitting and clustering, by varying group sizes. By grouping channels into 2, 4, and 8, it prioritizes deeper layer features, enhancing local semantic content and abstract feature representation. This methodological advancement significantly boosts classification accuracy, achieving an impressive 92.88% on Prostate Gleason dataset, outperforming other CNN models. To demonstrate the generalizability of ProsGradNet over different datasets, experiments are performed on Kasturba Medical College (KMC) Kidney dataset as well. The results further confirm the superiority of the proposed ProsGradNet model, with a classification accuracy of 92.68% on the KMC Kidney dataset. This demonstrates the model's potential to be applied effectively across various histopathological datasets, making it a valuable tool to fight against cancer.

1. Introduction

Prostate cancer, a widespread malignancy among men, is a major health concern due to its high incidence and significant mortality risk. The World Health Organization (WHO) reported approximately 1.4 million new cases of prostate cancer in 2020 [1]. The Lancet Commission on PCa [2] later estimates that this incidence of prostate cancer will eventually double by 2040, reaching a staggering 2.9 million cases annually. This anticipated rise in the number of cases poses significant challenges for affected individuals, healthcare systems, and clinicians. It primarily affects older men, with the likelihood of diagnosis increasing significantly over the age of 50. As early stage prostate cancer is often treatable with high survival rate, timely and accurate detection is very crucial for effective treatment and management. However, if left undiagnosed, it can progress to more critical stages leading to morbidity and mortality. The Gleason grading system [3] for PCa, developed by Donald Gleason in 1966, has undergone significant changes since

its inception. Recently, it was revised at a 2005 consensus conference held by international experts in urological pathology, organized by the International Society of Urological Pathology (ISUP) [4] in which the Gleason scores [5] were simplified into meaningful categories. Gleason scores typically range from 2 to 10. However, because pathologists found little distinction between scores 2 to 5, the practical range is from 6 to 10, with 6 indicating the lowest grade. Grade 1 includes Gleason grades of 6 or lower, characterized by well differentiated cells and well defined glandular structures that closely resembles normal prostate tissues. Grade 2, with a Gleason score of 3+4 = 7, consists of moderately differentiated cells. While most of the cells form well-defined glands similar to those in Grade 1, there is a minor component of poorly formed or fused glands in this grade. Grade 3 covers a Gleason score of 4+3 = 7, and includes cells that are poorly to moderately differentiated. These cells show greater irregularity in glandular structure, significant cellular atypia, and less resemblance to normal prostate tissue. Grade

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4, with a Gleason score of 8, comprises of poorly differentiated cells exhibiting a high degree of abnormality. The glandular architecture is significantly disrupted, with marked variation in cell size and shape. Grade 5 comprises of Gleason scores ranging from 9 to 10, whose cells are undifferentiated and anaplastic, showing a very high grade. They lack glandular formation and often appear as solid sheets, cords, or single cells. The architecture is highly disorganized, with extreme cellular atypia and high mitotic activity, showing very little resemblance to normal prostate tissue [6].

Prostate cancer diagnosis involves collecting clinical data, including patient history, conducting biomarker tests like PSA levels [7] and using medical imaging techniques such as MRI and TRUS [8] for visualizing the prostate gland to identify any abnormalities present. Biopsies, guided by these imaging techniques yield tissue samples that pathologists examine under a microscope to produce histopathology images. These images confirm the presence of cancer, assess its grade (how abnormal the cancer cell appears under a microscope compared to normal cells), and stage (extent of spread), which is crucial for treatment planning and prognosis determination. Manual methods like Digital Rectal Exams (DRE) and biopsies depends on the ability of the physicians to detect abnormalities, which may result in missing small or hard-to-reach tumors. The invasive nature of biopsies often leads to significant discomfort, potential health risks, and is time consuming. The evolution of machine learning and Artificial Neural Networks (ANNs) [9] provides a more accurate approach for detecting PCa by identifying subtle patterns and abnormalities that may be missed by the human eye. ANNs can quickly analyze imaging data and provide rapid results that accelerate the diagnostic process and reduce the waiting time for patients.

2. Related studies

Traditional machine learning involves extracting features like texture patterns using techniques such as histogram analysis, GLCM etc. These features are then fed into classifiers like Support Vector Machines (SVM) [9], Random Forest [10], Gradient Boosting Machines (GBM) [11] to distinguish the cancer types or to effectively predict the malignancy.

Recently, the emergence of deep learning (DL) techniques [12] has dramatically transformed the medical field, providing substantial improvements over traditional methods. However, the field is transitioning toward automatic DL methods, which aim to automate feature extraction and classification, potentially enhancing scalability and performance across diverse datasets and imaging modalities. As the traditional methods typically rely on manual feature engineering [13] and includes separate processes for classification [14], DL integrates these steps into a single framework. This automation allows DL models, especially Convolutional Neural Networks (CNNs) [15] to learn complex patterns directly from raw data, leading to improved performance and scalability. Studies on prostate cancer classification predominantly use MRI based images rather than histopathological images. XmasNet [16], developed by Liu has demonstrated superior performance in 2017 PROSTATEx challenge [17] showcasing its capability to differentiate benign and malignant prostate lesions from mpMRI images using semantic learning machine algorithm. The advent of CNN models was pioneered by some foundational architectures like VGG16 [18], AlexNet [19], Resnet50 [18], InceptionResnetV2 [20] and so on, which laid the foundation of development of more advanced architectures using the power of deep convolutional layers [21]. But, all these foundational architectures have a larger number of layers, which in turn increases the computational complexity of the model. The integration of attention mechanisms like Squeeze and Excitation (SE) [22] into CNN architectures has significantly enhanced prediction accuracy while maintaining computational efficiency. SqueezeNet [23], COVIDDiagnosisNet [24], FE-BkCapsNet [25] and TurkerNet [26] are

some of the architectures that demonstrates the capability of this mechanism to improve the performance of DL models in various medical classification tasks. Compared to the 60 million, 138 million, and 25 million parameters of AlexNet, VGG16, and ResNet50, respectively, SqueezeNet and TurkerNet has relatively smaller number of trainable parameters, ranging around 1.2 million and 1.6 million, respectively. Notably, TurkerNet has demonstrated strong performance in the binary classification of skin cancer images. Recent studies such as BHCNet [27] and BreastNet [28], that focused on the binary classification of the BreakHis dataset into benign and malignant classes have effectively utilized the SE block to model channel-wise relationships. Furthermore, the Channel Block Attention Module (CBAM) employed in models such as LiverNet [29,30], BreastNet [28] incorporates Atrous Spatial Pyramid Pooling (ASPP) mechanism, enhancing the feature extraction process. Unlike SE blocks, which focuses on enhancing channelwise features, the ASPP block captures multiscale contextual features through dilated convolutions. The LiverNet model performed well on two different datasets, namely TCGA liver and KMC liver datasets, while maintaining a lower number of parameters. However, a significant limitation of this study is the relatively small size and limited variance of the dataset used. Additionally, the performance of the BreastNet architecture was evaluated solely on the BreakHis dataset. In recent years, the introduction of shuffling mechanism has marked a significant advancement in the field of medical imaging. One prominent example is the ShuffleNet [31,32] architecture, which utilizes channel shuffling operation for the flow of information within the groups. Even though the model has shown good performance in the classification of cancerous images, the model complexity is very high with around 23 million parameters and 9 Giga Floating Point Operations (FLOPS). The efficacy of ShuffleNet has been used in various applications, including medical image segmentation, tumor classification, COVID-19 detection etc. In particular, models like ShuffleSwishNet-108 [33] is build on this concept by integrating additional mechanisms such as the Swish activation function, which outperforms the ReLU activation function used in basic shuffleNet model. Even though the model performs better than ShuffleNet, there is no much significant decrease in the number of parameters. Building on these advancements, RCCGNet [34] represents a cutting edge CNN architecture for the automatic classification of Renal Cell Carcinoma (RCC) from histopathology images. The model was evaluated on a newly introduced KMC Kidney dataset [34], which achieved an accuracy of 90.14% with only 0.36 million parameters.

Considering the critical significance of cancer classification in healthcare, misclassifications (particularly false negatives and false positives) can lead to inappropriate treatment strategies, which may negatively impact the health of the patients. The proposed ProsGradNet model, therefore, focuses on delivering more accurate predictions along with reducing the model complexity. Drawing inspiration from the RCCGNet architecture, certain blocks were modified and new techniques were integrated to develop the proposed model. These modifications, including the introduction of a novel Context Guided Shared Channel Residual (CGSCR) block, have led to notable improvements in classification accuracy, demonstrating the robustness and effectiveness of the approach.

The primary contributions in this paper are as follows:

- (a) A novel CGSCR block has been introduced, offering a more structured approach for splitting and clustering of channels, leading to better feature representation.
- (b) A robust ProsGradNet model has been designed and developed for the classification of PCa into its respective subtypes from histopathology images.
- (c) The performance of the proposed ProsGradNet model is evaluated and compared against the most recent CNN models on two different histopathological datasets, demonstrating superior accuracy over all five comparison models.

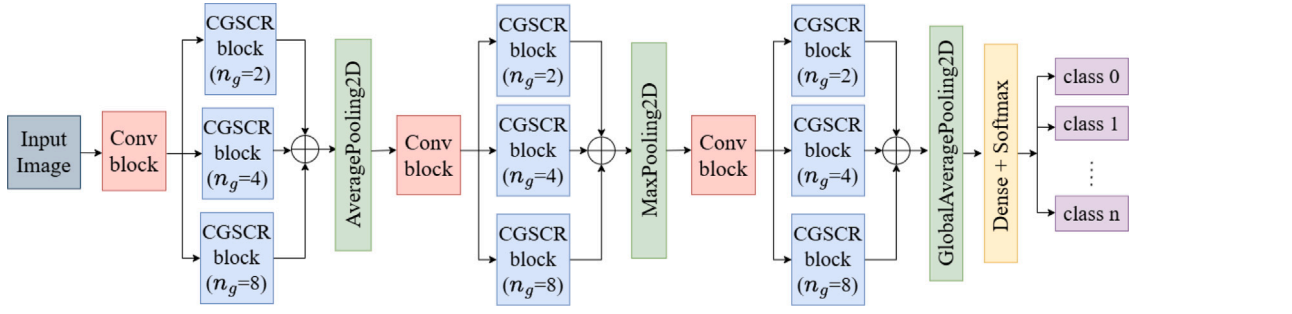


Fig. 1. Architecture of the proposed ProsGradNet model.

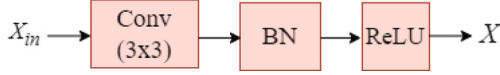


Fig. 2. Convolutional block.

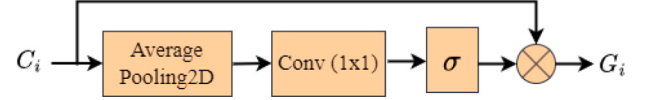


Fig. 3. Gating block [34].

3. Proposed model

This section details the architecture of the proposed ProsGradNet model. The architecture of the proposed model is illustrated as in Fig. 1. The proposed model consists of several convolutional layers and novel CGSCR blocks, that collectively leverages the feature representation and improves the classification accuracy. The proposed model extracts features through three stages, which are separated by a pooling layer, that progressively reduces the spatial dimensions. Each stage consists of three sets of parallel CGSCR blocks, followed by a convolutional block. The output features from these parallel paths are subsequently added, which forms the first stage. The pooling layer reduces the dimension of feature map to half using operations like max pooling and average pooling. This process is then repeated in the second and third stages, with the number of filters in the convolutional layers doubling at each stage. The output of the third stage is then globally pooled, providing the global content of each channel. This pooled output serves as the input to the classification layer, where the number of neurons is equal to the number of classes. A softmax activation function finally assigns probability scores to each class.

The proposed model takes an input image of shape (256,256,3), which corresponds to the height, width, and the number of channels, respectively. This input X_{in} is then passed to a simple convolutional block as shown in Fig. 2, which provides an initial feature map X with 16 channels as given in Eq. (1).

$$X = \text{ReLU} \left(\text{BN} \left(\text{Conv}_{3 \times 3} (X_{in}) \right) \right) \quad (1)$$

This output is fed to three sets of CGSCR blocks in a parallel fashion. The introduced CGSCR block applies gating mechanisms for each group which highlights the most significant channels. These blocks are characterized by the number of groups in them, which are 2, 4 and 8 respectively. The CGSCR block, which accepts lower layer features X as the input is passed through a convolutional block that extracts higher layer features Y , which is represented as in Eq. (2).

$$Y = \text{ReLU} \left(\text{BN} \left(\text{Conv}_{3 \times 3} (X) \right) \right) \quad (2)$$

The idea of group wise channel splitting and clustering is introduced at this stage, where the channels are split into n_g , where n_g represents the number of groups. Fig. 5 represents the CGSCR block with $n_g = 2, 4$, and 8 respectively. For a given $X = \{x^1, x^2, x^3, \dots, x^c\}$, and $n_g = 4$, the four different groups of outputs X_1, X_2, X_3 and X_4 after channel splitting are as given in Eqs. (3), (4), (5) and (6) respectively.

$$X_1 = (x^1, x^2, x^3, \dots, x^{c/4}) \quad (3)$$

$$X_2 = (x^{c/4+1}, x^{c/4+2}, x^{c/4+3}, \dots, x^{2c/4}) \quad (4)$$

$$X_3 = (x^{2c/4+1}, x^{2c/4+2}, x^{2c/4+3}, \dots, x^{3c/4}) \quad (5)$$

$$X_4 = (x^{3c/4+1}, x^{3c/4+2}, x^{3c/4+3}, \dots, x^c) \quad (6)$$

In a similar way, the higher order features Y are also split into Y_1, Y_2, Y_3 and Y_4 respectively. After splitting, new clusters of features C_1, C_2, C_3 and C_4 are created from different groups of X and Y . Eq. (7), (8), (9) and (10) represents the clusters formed after feature concatenation.

$$C_1 = X_1 \odot Y_2 \odot Y_3 \odot Y_4 \quad (7)$$

$$C_2 = Y_1 \odot X_2 \odot Y_3 \odot Y_4 \quad (8)$$

$$C_3 = Y_1 \odot Y_2 \odot X_3 \odot Y_4 \quad (9)$$

$$C_4 = Y_1 \odot Y_2 \odot Y_3 \odot X_4 \quad (10)$$

This is a more explicit and structured approach for group wise feature concatenation. By varying the group size, the CGSCR block can capture abstract features by mixing both higher and lower order features. The distribution of the features of X and Y plays an important role while clustering the channels. Suppose that the channels are grouped into n , then, one channel from the first group (X) will be collectively clustered with the remaining $n-1$ channels from the second group (Y), thereby prioritizing the extraction of deeper layer features. In the SCR block of RCCGNet [34], the channels were only divided into 2. This results in the extraction of 50% of lower level features and 50% of higher level features. But in the CGSCR block, channels are grouped into 2, 4 and 8 respectively. By keeping the group size as 4, 25% of lower level features and 75% of higher level features gets extracted. In a similar fashion, dividing the channels into a group of 8 channels favors in the distribution of 87.5% of higher level features and remaining 12.5% of lower level features. The clustered features contain most of the information from the higher level and a part of information from lower layers. This lower level information can be treated as a residual connection, thereby preserving the information from the preceding layers. Moreover, this method of splitting the channels into varying groups give the advantage of extracting more abstract features, thereby strengthening the local semantic content.

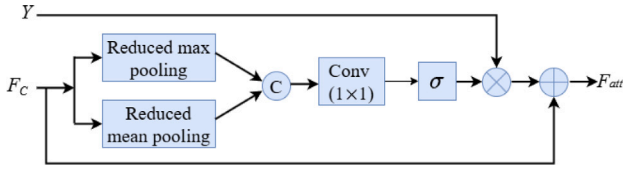


Fig. 4. Context guided block [35].

A gating method [34] as used in RCCGNet is applied to each of the concatenated outputs C_i and is given in Fig. 3. It consists of an average pooling layer with kernel size = $(W \times H)$, where W and H represents the width and height of the C_i . Following that, a sigmoid activated convolutional layer computes the channel wise attention weights which indicate how important each channel is within the feature map. The gating mechanism thus helps in emphasizing attention on the most informative features. The G_i s obtained after applying the gating mechanisms as in Eq. (11) are then concatenated to produce F as given in Eq. (12).

$$G_i = \sigma \left(\text{Conv}_{(1 \times 1)} \left(\text{AvgPool} \left(C_i \right) \right) \right) \times C_i \quad (11)$$

$$F = G_1 \odot G_2 \odot G_3 \odot G_4 \quad (12)$$

Next, a pointwise convolution with ReLU activation is applied to reduce the number of filters of the aggregated features, resulting in a feature map F_C . This process will eventually introduce a nonlinearity. An additional context guided block [35] further enhances the feature representation of F_C . A pixelwise score is assigned to F_C according to their relevance, and it then updates higher level feature map with this score, which in turn provides an additional support to F_C . The score $F_{C_{sig}}$ is achieved by computing the mean $F_{C_{avg}}$ and maximum $F_{C_{max}}$ of feature F_C along the channel axis, concatenating, and applying a sigmoid activated convolutional layer. By multiplying this attention map with higher order features Y , and adding them back to F_C , the model can integrate the contextually enhanced features of Y directly into F_C to produce F_{att} . The block diagram of the Context Guided block is given in Fig. 4 and the corresponding mathematical representations are given in Eqs. (13), (14), (15) and (16) respectively. This F_{att} after passing through a convolutional layer processes the final output X_{out} of the CGSCR block.

$$F_{C_{avg}} = \text{Avg} \left(F_C \right)_{axis=3} \quad (13)$$

$$F_{C_{max}} = \text{Max} \left(F_C \right)_{axis=3} \quad (14)$$

$$F_{C_{sig}} = \text{Sigmoid} \left(\text{Conv}_{(1 \times 1)} \left(F_{C_{avg}} \odot F_{C_{max}} \right) \right) \quad (15)$$

$$F_{att} = F_{C_{sig}} \times Y + F_C \quad (16)$$

4. Training and implementation

This section includes a comprehensive overview of the datasets used, experimental setup employed for training the models, hyperparameter settings, training procedures and evaluation metrics to assess the model performance.

4.1. Dataset

The proposed ProsGradNet is evaluated using two well known histopathological datasets, namely, Prostate Gleason dataset, detailed in [36] and KMC Kidney dataset, described in [34].

Table 1

Distribution of training, test, and validation patches across different classes in the Prostate Gleason dataset used in this study.

	Classes	Training patches	Test patches	Validation patches
Original Dataset	Class 0	11 200	2800	2240
	Class 1	11 200	2800	2240
	Class 2	7936	1985	1590
	Class 3	10 814	2704	2163
	Total	41,150	10,289	8,233
Custom Dataset	Class 0	3018	754	800
	Class 1	3000	700	728
	Class 2	2851	700	718
	Class 3	2815	700	693
	Total	11,684	2,854	2,939

Table 2

Distribution of training, test, and validation patches across different grades in the KMC Kidney dataset.

Classes	Training patches	Test patches	Validation patches
Grade 0	693	102	102
Grade 1	708	107	104
Grade 2	648	96	96
Grade 3	735	105	105
Grade 4	648	96	96
Total	3432	506	503

1. Prostate Gleason Dataset : It is a collection of prostate cancer biopsy images that are sourced from the PANDA dataset [36]. It comprises of approximately 70,000 image patches, each of size 256×256 pixels. These patches are categorized into four classes based on their Gleason score/grade as class 0 (Gleason score cannot be determined), class 1 (Gleason score ≤ 6), class 2 (Gleason score = 7), and class 3 (Gleason score ≥ 8 , 9, 10). A total of 41,150 training images, 10,289 testing images and 8233 validating images are present in the original dataset. Due to resource limitation, a more balanced custom dataset is created from the resampled image patches of the original one. In this study, from the resampled dataset, 11,684, 2854, and 2939 patches were randomly selected for training, testing and validating the dataset respectively. Table 1 shows the distribution of image patches in the Prostate Gleason dataset and Fig. 6 presents the four different Gleason patterns that are present in the dataset.
2. KMC Kidney dataset : In order to enhance the generalization ability of the model, the proposed model is evaluated on KMC Kidney dataset, collected from Kasturba Medical College (KMC) Mangalore, Manipal Academy of Higher Education (MAHE), Manipal, Karnataka, India [34]. Fig. 7 shows the five different classes of Renal Cell Carcinoma contained in the dataset. Table 2 shows the distribution of the KMC Kidney dataset over the five classes with a total of 4441 patches, out of which, 3432 patches were used for training, 506 patches for testing and 503 patches for validating the model.

4.2. Training setup

The proposed model was trained on a workstation equipped with an Intel (R) Xenon (R) W-2235 CPU, 32 GB of RAM, and an NVIDIA Quadro RTX4000 GPU with 8 GB of onboard memory. The system is configured with Windows 10 Pro as the operating system and utilized

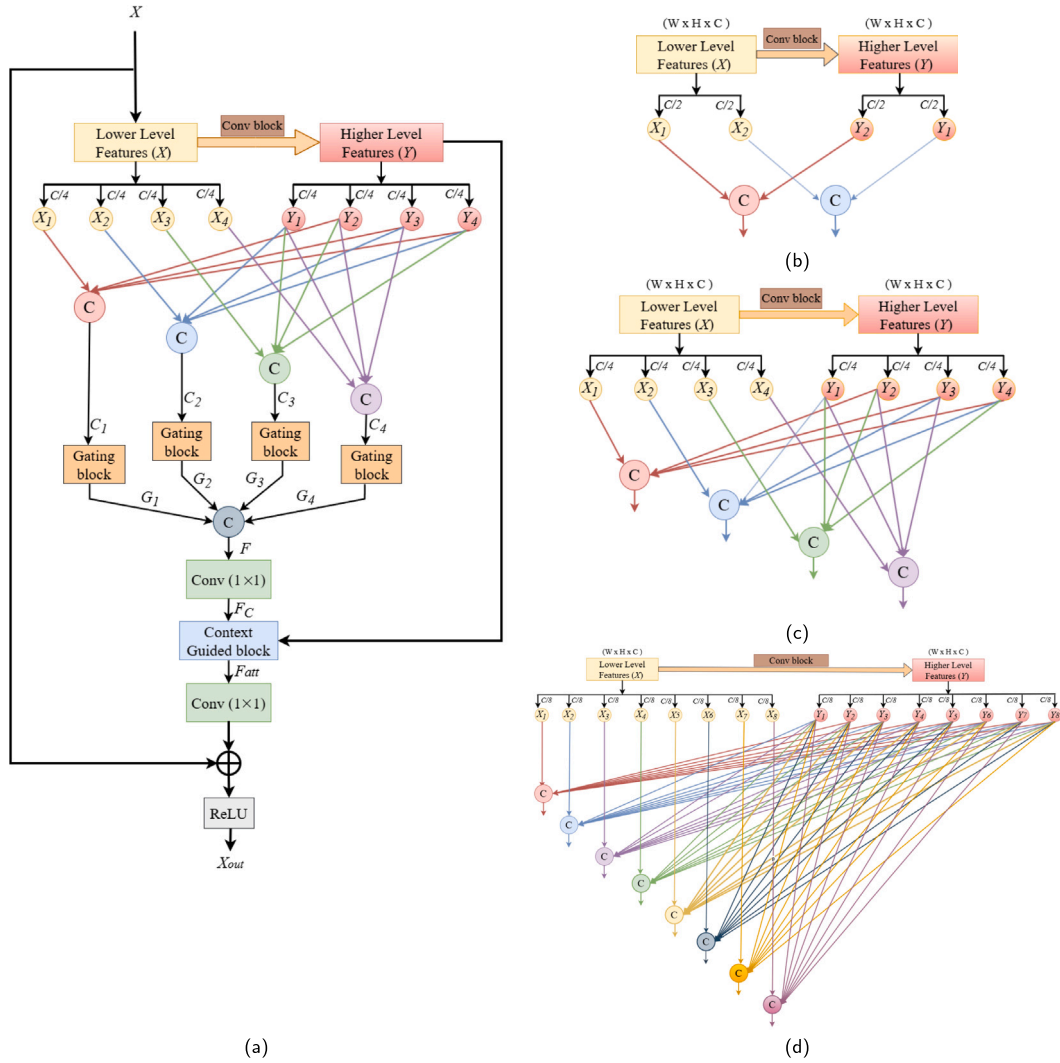


Fig. 5. (a) Context Guided Shared Channel Residual Block (b) Channel splitting with $n_g=2$ (c) Channel splitting with $n_g=4$ (d) Channel splitting with $n_g=8$.

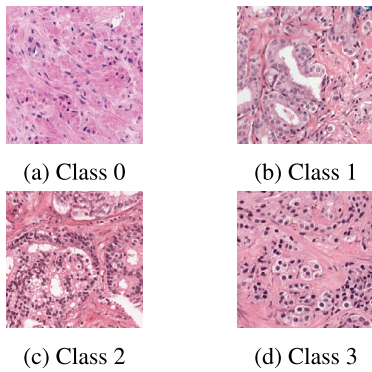


Fig. 6. Visualization of the four different classes of Prostate Gleason dataset.

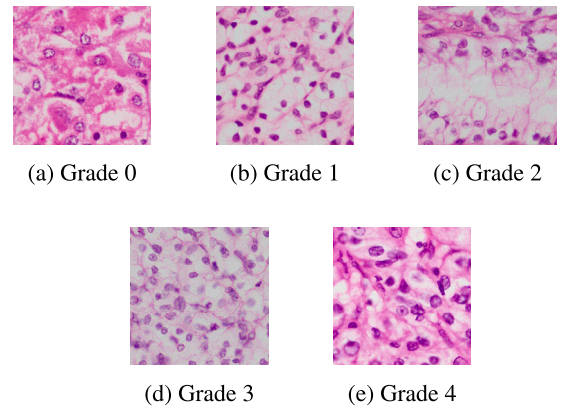


Fig. 7. Visualization of the five different grades of KMC Kidney dataset.

Tensorflow 2.5.0 for the implementation. The Adam optimizer is employed here with an initial learning rate set to 0.001. Further, a learning rate scheduler is used to change the learning rate as the training progresses. The proposed model and all five comparison models were trained for 65 epochs in both the datasets with a batch size of 4. The

loss function used here is Categorical Cross Entropy [34]. The performance of all the models were evaluated using the four most common evaluation metrics namely Accuracy [9], Precision [5], Recall [9], and F1 Score [9]. The source code of the proposed ProsGradNet will be available at: <https://github.com/shyamfec/ProsGradNet>.

Table 3
Ablation study of proposed ProsGradNet on Prostate Gleason dataset.

Model	Context guided block	$n_g = 2$	$n_g = 4$	$n_g = 8$	Acc	Pr	Re	F1
Intermediate Model 1	×	✓	✓	✓	0.9171	0.9176	0.9172	0.9172
Intermediate model 2	✓	✓	×	×	0.9103	0.9112	0.9107	0.9109
Intermediate model 3	✓	×	✓	×	0.9162	0.9165	0.9166	0.9166
Intermediate Model 4	✓	×	×	✓	0.9145	0.9164	0.9148	0.9154
Proposed ProsGradNet	✓	✓	✓	✓	0.9288	0.9291	0.9293	0.9292

Table 4
Ablation study of proposed ProsGradNet on KMC Kidney dataset.

Model	Context guided block	$n_g = 2$	$n_g = 4$	$n_g = 8$	Acc	Pr	Re	F1
Intermediate Model 1	×	✓	✓	✓	0.8754	0.8788	0.8742	0.8756
Intermediate model 2	✓	✓	×	×	0.8774	0.8772	0.8765	0.8750
Intermediate model 3	✓	×	✓	×	0.8754	0.8741	0.8756	0.8719
Intermediate Model 4	✓	×	×	✓	0.8754	0.8843	0.8767	0.8770
Proposed ProsGradNet	✓	✓	✓	✓	0.9268	0.9276	0.9273	0.9263

5. Results and discussion

This section presents the results of the ablation study, followed by the results of the comparative studies, along with a detailed discussion of their implications. It also includes an analysis of the computational complexity of each model using various metrics. Furthermore, boxplots are used to visually represent key metrics across the two datasets. Through these analysis, the strengths and limitations of the proposed model are critically evaluated, with the best performing results highlighted using bold text.

5.1. Ablation study

The proposed ProsGradNet model features a CGSCR block, which has three different channel grouping mechanisms and a context guided block. In order to understand the contribution of each component toward the overall performance of the model, ablation study was conducted by systematically removing one component at a time to create different versions of the model. Each of them should be identical to the baseline except for the specific component being analyzed. Here, four different intermediate models were created by modifying the components of the proposed model. Each intermediate model represents a variant, where specific components and grouping strategies were altered. Table 3 and Table 4 presents the results of the ablation study conducted on the proposed model on both the datasets respectively.

1. Intermediate model 1: Intermediate model 1 involves the removal of context guided block from the architecture. It gives an accuracy of 91.71% on Prostate Gleason dataset. However, on the KMC kidney dataset, there is a 5% decrease across all four evaluation metrics compared to the proposed model. This clearly suggests that the context guided block significantly influences the model performance.
2. Intermediate model 2: Intermediate model 2 involves splitting and clustering the channels using a group size of 2 ($gr = 2$) exclusively. Splitting of channels into 2 alone gives an accuracy of 91.03% and 87.74% on prostate and kidney datasets respectively, both lower than the proposed model, indicating that this configuration is less effective.
3. Intermediate model 3: Intermediate model 3 exclusively splits and clusters channels using a group size of 4 ($gr = 4$). This configuration gives an accuracy of 91.62% on Prostate Gleason dataset, which still falls short of the proposed model. This variant gives a 5% decrease in accuracy on kidney dataset.
4. Intermediate model 4: Intermediate model 4 involves splitting and clustering the channels using a group size of 8 ($gr = 8$) without utilizing $gr = 2$ and $gr = 4$. Splitting with $gr = 8$ gives an

accuracy of 91.45% and 87.54% on prostate and kidney datasets respectively, which are also lower than the proposed model.

The proposed model thus consistently outperforms the intermediate models, suggesting that the combination of components and strategies it employs is optimal. The performance variations across intermediate models highlights the significance of the specific design choices, such as the context-guided block, which helps in achieving high performance for the KMC Kidney dataset.

5.2. Comparison with benchmark models

Table 5 presents a detailed assessment of the performance of different CNN models on the Prostate Gleason dataset, employing four key evaluation metrics: Accuracy, F1-score, Precision, and Recall. These metrics evaluates how effectively each model classifies PCa into distinct classes. The models under evaluation include ShuffleNet [31], BreastNet [28], LiverNet [30], RCCGNet [34], TurkerNet [26], and the proposed ProsGradNet. According to the results provided in the table, the proposed ProsGradNet outperforms all other models across all metrics, achieving an impressive overall accuracy of 92.88%. Specifically, the Precision, Recall, and F1-score for the ProsGradNet model are recorded as 92.91%, 92.93%, and 92.92%, respectively. Among the comparison models, ShuffleNet demonstrates the strongest performance after ProsGradNet, with an accuracy of 90.74%. This model achieved 90.75% F1-score, 90.68% Recall, and 91.05% Precision. ShuffleNet has the best Precision for Class 0 and Class 1 with values 93.69% and 87.63% respectively. It also has the highest Recall value for Class 0 with 93.89%. However, its overall Precision and Recall are lower than the proposed model. In contrast, BreastNet exhibits comparatively weaker results with values of 86.79%, 86.8%, 86.87%, and 86.79% for Accuracy, F1-score, Recall, and Precision, respectively. The RCCGNet also performs well among the comparison model giving an overall accuracy of 89.83% and it excels in Precision for Class 2 (94.45%). The most recent among the benchmarks, TurkerNet demonstrates strong results with competitive scores across all metrics. It achieves notable Precision in Class 3 (99.57%) and Recall in Class 0 (90.71%), but the proposed model still outperforms it in overall metrics. These results highlight the superior performance of the proposed ProsGradNet in classifying PCa accurately when compared to existing models. The significant margins in Accuracy, Precision, Recall, and F1-score underscore the effectiveness of the proposed model in handling the complexities of the Prostate Gleason dataset.

A similar comparative analysis was conducted on KMC Kidney dataset as given in Table 6, in which the proposed ProsGradNet achieved the best results with 92.68% Accuracy, 92.63% F1-score, 92.73% Recall, and 92.76% Precision respectively. The RCCGNet shows

Table 5
Performance evaluation of proposed and other existing DL models on Prostate Gleason dataset.

Metrics	Class	ShuffleNet (2018)	BreastNet (2020)	LiverNet (2021)	RCCGNet (2023)	TurkerNet (2024)	Proposed model
Precision	0	0.9369	0.8401	0.8631	0.8462	0.8571	0.9018
	1	0.8763	0.7557	0.7779	0.8147	0.8408	0.8726
	2	0.9351	0.8832	0.9416	0.9445	0.9287	0.9421
	3	0.9900	0.9929	1.0	0.9985	0.9957	1.0
	Overall	0.9105	0.8679	0.8956	0.9010	0.9056	0.9291
Recall	0	0.9389	0.8222	0.8700	0.9124	0.9071	0.9018
	1	0.81	0.7557	0.8257	0.8042	0.8	0.8614
	2	0.8857	0.8971	0.8757	0.8757	0.9128	0.9542
	3	0.9928	1.0	1.0	1.0	0.9985	1.0
	Overall	0.9068	0.8687	0.8928	0.8981	0.9046	0.9293
F1-score	0	0.8872	0.8310	0.8665	0.8781	0.8814	0.9018
	1	0.8418	0.7557	0.8011	0.8094	0.8199	0.8670
	2	0.9097	0.8901	0.9074	0.9088	0.9207	0.9481
	3	0.9914	0.9964	1.0	0.9992	0.9971	1.0
	Overall	0.9075	0.8683	0.8937	0.8989	0.9048	0.9292
Accuracy	0	0.9369	0.9117	0.9292	0.9330	0.9355	0.9481
	1	0.9253	0.8801	0.8994	0.9071	0.9138	0.9351
	2	0.9569	0.9456	0.9562	0.9569	0.9614	0.9744
	3	0.9957	0.9982	1.0	0.9996	0.9985	1.0
	Overall	0.9074	0.8679	0.8924	0.8983	0.9046	0.9288

Table 6
Performance evaluation of proposed and other existing DL models on KMC Kidney dataset with existing models.

Metrics	Class	ShuffleNet (2018)	BreastNet (2020)	LiverNet (2021)	RCCGNet (2023)	TurkerNet (2024)	Proposed Model
Precision	0	0.9901	1.0	0.9166	0.9756	0.9902	0.9805
	1	0.9166	0.9122	0.9345	0.8846	0.9099	1.0
	2	0.7523	0.6868	0.6730	0.9047	0.7422	0.8736
	3	0.8217	0.7789	0.8210	0.7241	0.8173	0.8878
	4	0.9222	0.9062	0.9673	1.0	0.9450	0.8962
	Overall	0.8806	0.8313	0.8625	0.8978	0.8809	0.9276
Recall	0	0.9901	1.0	0.9707	0.9756	1.0	0.9901
	1	0.9252	0.9719	0.9345	0.8518	0.9439	0.8878
	2	0.8229	0.7083	0.7291	0.76	0.75	0.8645
	3	0.7904	0.7047	0.7428	1.0	0.8095	0.9047
	4	0.8645	0.9062	0.9270	0.8928	0.8958	0.9895
	Overall	0.8786	0.8582	0.8608	0.8961	0.8798	0.9273
F1-score	0	0.9901	1.0	0.9428	0.9756	0.9951	0.9853
	1	0.9209	0.9411	0.9345	0.8679	0.9266	0.9405
	2	0.7860	0.6974	0.7	0.8260	0.7461	0.8691
	3	0.8058	0.74	0.78	0.84	0.8133	0.8962
	4	0.8924	0.9062	0.9468	0.9433	0.9197	0.9405
	Overall	0.8790	0.8569	0.8608	0.8906	0.8802	0.9263
Accuracy	0	0.9960	1.0	0.9762	0.9859	0.9980	0.9940
	1	0.9664	0.9743	0.9723	0.9507	0.9683	0.9762
	2	0.9150	0.8833	0.8814	0.9436	0.9031	0.9505
	3	0.9209	0.8972	0.9130	0.9436	0.9229	0.9565
	4	0.9604	0.9644	0.9802	0.9788	0.9703	0.9762
	Overall	0.8794	0.8596	0.8616	0.9014	0.8814	0.9268

a competitive performance across all metrics with an Accuracy of 90.14%, F1-score of 89.06%, Recall of 89.61%, and Precision of 89.78%. The BreastNet model achieves perfect performance for Class 0 across all metrics, with each value being 100%. It also gives the best F1-score (94.11%) and Recall (97.19%) for Class 1. LiverNet shows strong F1-score (94.68%) and Accuracy (98.02%) for Class 4, but not for other classes. The poorest result was given by BreastNet with 85.96% Accuracy and 85.69% F1-score. The ShuffleNet was only able to give an average accuracy of 87.94% on this dataset. The enhanced performance of the proposed ProsGradNet model is contributed by the CGSCR block. The various group sizes allow the CGSCR block to effectively utilize the contextual information from each of the higher level and lower level feature maps. By utilizing the weights at both pixel and global levels, the CGSCR block will be able to enhance the understanding of local and global content. The proposed model thus consistently shows strong performance across different datasets, highlighting its ability to deliver superior results independent of the dataset.

Confusion matrix is another tool to evaluate the performance of the models. It helps in understanding how well the model is performing, which classes are correctly identified, and which classes are misclassified. Fig. 8 and Fig. 9 shows the confusion matrix of the proposed ProsGradNet model on Prostate Gleason dataset and KMC Kidney dataset respectively. For Prostate Gleason dataset, the ProsGradNet model correctly predicted 2651, out of a total of 2854 instances. The proposed model gave 700 correct predictions without any misclassifications for Class 3. The absence of misclassifications indicate that the features relevant to Class 3 are well captured by the model. It also maintains competitive performance for Class 0, Class 1 and Class 2 compared to all other comparison models. Similarly, the ProsGradNet model predicted 469, out of 506 instances for KMC Kidney dataset. Grade 0 and Grade 4 has the minimum number of misclassifications among all the five grades. Overall, the results indicate the robustness of the proposed ProsGradNet model in distinguishing between different classes and its effectiveness in minimizing misclassifications. The presence of misclassifications suggests that there are some features which

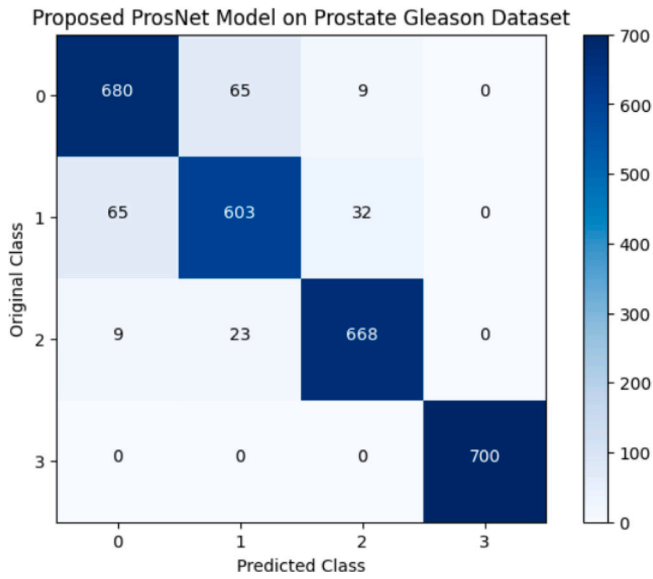


Fig. 8. Confusion matrix of proposed ProsGradNet on Prostate Gleason dataset [35].

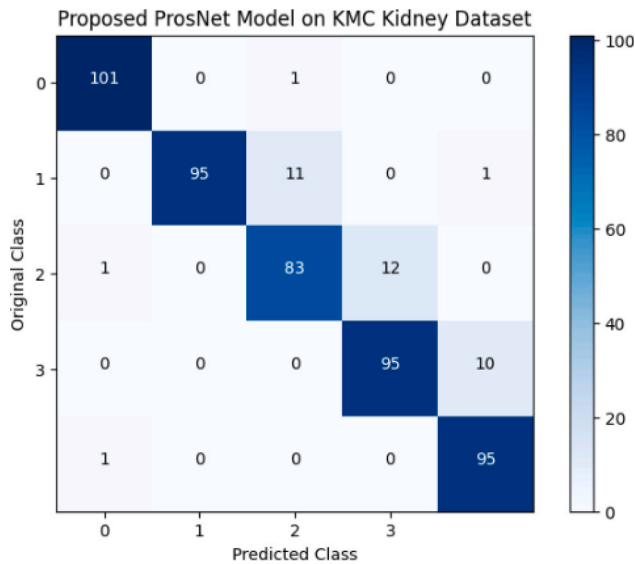


Fig. 9. Confusion matrix of proposed ProsGradNet on KMC Kidney dataset.

are similar among these classes that leads to confusion.

5.3. Computational complexity analysis

The computational complexity of DL models refers to the amount of computational resources required to train and run these models. In Table 7 and Table 8, several models are compared with the proposed model based on their computational complexity using various metrics on the Prostate Gleason and KMC Kidney dataset respectively. These metrics includes the number of parameters, floating-point operations per second (FLOPS), training time, and test time per sample. The proposed model has the lowest parameter counts on both the datasets, indicating a relatively smaller and potentially less complex architecture compared to others like ShuffleNet (23.82 M) and TurkerNet (1.59 M). FLOPS measure the number of floating point operations required to complete a forward pass. Lower FLOPS indicate less computational demand. The proposed model, with 4.87 G FLOPS on Prostate Gleason dataset, suggests a moderate computational load, which is lower than

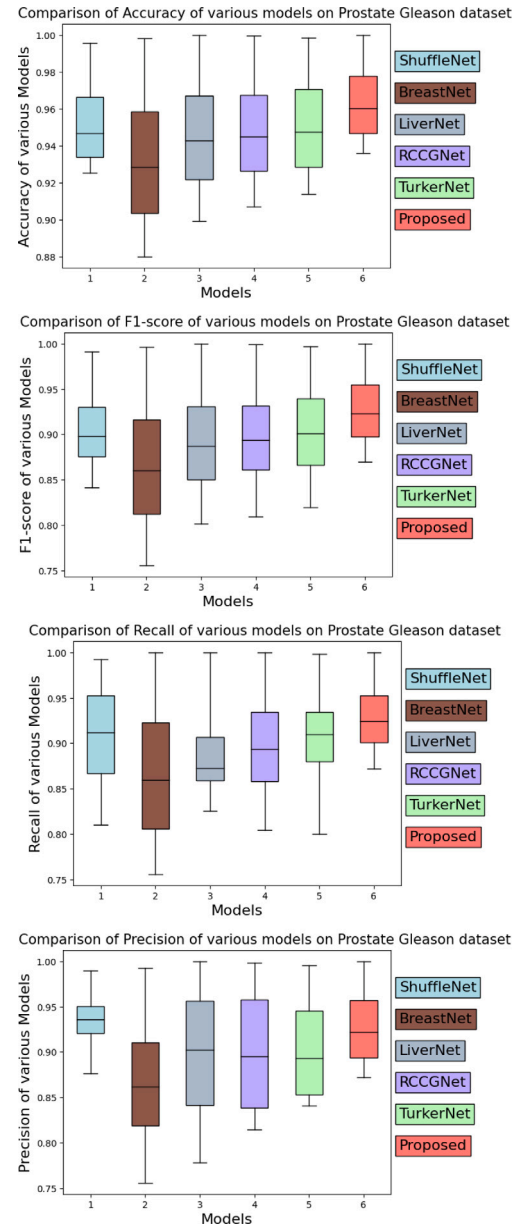


Fig. 10. Comparison of different DL models on Prostate Gleason dataset using boxplots.

that of ShuffleNet (8.69 G), BreastNet (7.39 G), RCCGNet (5.82 G), but higher than TurkerNet (0.458 G). The number of FLOPS for KMC Kidney dataset is 3.73 G, which is also lower than all other comparison models except TurkerNet (0.458 G). Despite having fewer parameters and lower FLOPS compared to models like ShuffleNet and RCCGNet, the test time per sample for the proposed ProsGradNet is higher. This increased test time is due to the operations within the CGSCR block that involves multiple conditional statements and loops.

5.4. Statistical analysis using boxplots

Boxplots helps to visualize the distribution of performance metrics across different classes. The boxplot shown in Fig. 10 and Fig. 11 compares the Accuracy, F1-score, Recall and Precision of various models on both the datasets. The line inside the interquartile range (IQR) of the boxplot represents the median of the dataset. The median values of Accuracy, F1-score and Recall are higher for the proposed model compared to other models in Prostate Gleason dataset. Also, the box

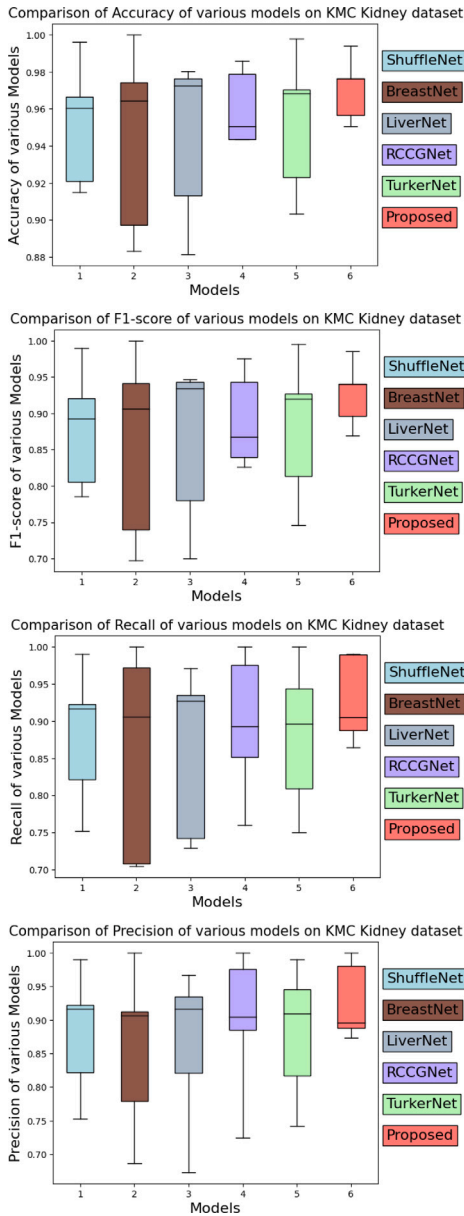


Fig. 11. Comparison of different DL models on KMC Kidney dataset using boxplots.

Table 7

Computational complexity of various models on Prostate Gleason dataset.

Model	Parameters (in M)	FLOPS (in G)	Test time per sample (in ms)
ShuffleNet (2018)	23.82 M	8.69 G	21
BreastNet (2020)	0.6058 M	7.39 G	13
LiverNet (2021)	0.6051 M	4.97 G	21
RCCGNet (2023)	0.3649 M	5.82 G	36
TurkerNet (2024)	1.59 M	0.458 G	10
Proposed ProsGradNet	0.340 M	4.87 G	34

representing IQR is narrow in these cases. This indicates that the middle 50% of the data points are close to each other, suggesting consistency in the performance of the ProsGradNet. In the same dataset, the median value of Precision is highest for ShuffleNet, while BreastNet has the lowest median value. For the KMC Kidney dataset, the median values of accuracy and F1-score is highest for the ProsGradNet model and the IQR is also narrow compared to other models. The median value of Precision and Recall is highest for ShuffleNet in the same dataset.

Table 8

Computational complexity of various models on KMC Kidney dataset.

Model	Parameters (in M)	FLOPS (in G)	Test time per sample (in ms)
ShuffleNet (2018)	23.83 M	6.65 G	19
BreastNet (2020)	0.6061 M	5.66 G	11
LiverNet (2021)	0.6053 M	3.81 G	19
RCCGNet (2023)	0.3651 M	4.46 G	15
TurkerNet (2024)	1.59 M	0.458 G	11
Proposed ProsGradNet	0.340 M	3.73 G	29

6. Conclusions

This paper highlighted the newly proposed ProsGradNet model for the classification of prostate cancer grades from histopathology images. The newly introduced CGSCR block in the proposed ProsGradNet model significantly boosted feature representation. The proposed ProsGradNet model strategically integrated CGSCR blocks across three stages of feature extraction, each stage progressively refining spatial dimensions and leveraging varying group sizes (2, 4, and 8) to capture both lower and higher-level features. The inclusion of a gating mechanism within CGSCR block enhances feature selection by emphasizing the most informative channels, thereby enriching the model's ability to recognize subtle histopathological differences. Moreover, the context-guided block refines feature maps, ensuring that the model optimally integrates contextual information to improve diagnostic accuracy. The proposed model gave an average accuracy of 92.85% on Prostate Gleason dataset and 92.68% on KMC Kidney dataset. These architectural refinements highlights its adaptability across diverse histopathological datasets, making it a promising tool for advancing prostate cancer diagnosis and treatment strategies. The major limitation of the model is its higher inference time. The future work will be focused on optimizing the model to reduce the test time without compromising the model performance.

CRedit authorship contribution statement

Akshaya Prabhu: Writing – original draft, Methodology, Investigation, Conceptualization. **Sravya Nedungatt:** Writing – review & editing, Conceptualization. **Shyam Lal:** Writing – review & editing, Supervision, Conceptualization. **Jyoti Kini:** Writing – review & editing, Validation, Resources, Formal analysis, Data curation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The source code of the proposed ProsGradNet and the dataset will be available at: <https://github.com/shyamfec/ProsGradNet>.

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